Group and solo flights

Technical report on the FFVL cross-country archive

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22 October 2026 • analysis: `xc-lab/analysis/groups/group-identification` • grouping run `d120_eh500_ez300_th45_g120_q0.1`

Question. Do paragliders move differently when they fly in a group, and is it the group or the ground, day and pilots? **Data.** FFVL archive, 154 598 paraglider flights, 38.1-min contacts grouped by a null model (21.7% fly in a group). **Method.** Group time (G), solo flights (S) and the same flights alone (A) compared over *every lag* of the TA-MSD and on phase statistics, inside strata (zone, month, hour, site-day, flight) with 5–95% block-bootstrap bands.

Findings.

- **Straighter progress over 1000 s in a group:** $\Delta H_{\mathrm{loc}}(1000) = +0.055$ raw, $+0.037$ within the flight, $+0.043$ at the same site-day and hour (selection only $+0.003$). The gap opens above 150 s; nothing within one glide or climb.
- **It is not the zone:** reweighting solo flights to the cells, months and hours where groups fly removes at most 19% of the gap; H: G 0.916, S 0.899 $\rightarrow$ 0.904 at the same site-day and hour.
- **It follows the group in time:** on the same flights, $\Delta H_{\mathrm{loc}}(1000)$ is $+0.027$ entering and $+0.031$ leaving a group of at least 50 min, against -0.002 between consecutive windows with no change of state; before and after the group are alike (-0.003). Glides are faster and flatter in the group, in both directions.
- **More time climbing, more successful searches, faster and flatter glides, higher climb rate and cross-country speed** once stratified (the pooled readings of climb rate and speed reversed: composition).
- **Size:** most of the effect is there with 2 pilots; at the same place and hour only 7+ stands apart.
- **Leader and follower:** the pilot ahead flies a little faster but finds no more thermals. What groups are, how they form and break and who leads: report *Group structure*.

Limits. Strata fix pilot, day, site, hour and zone, and the before/during/after test the flight itself, not the moment pilots choose to fly together; contact = shared air within sight; phases on 1 Hz flights only.

1 Setup

1.1 Data and grouping

- **Flights.** 156 391 paraglider and 6 094 hang-glider flights on 6 209 days, cleaned tracks of the old repository (soaring, tag `legacy-final`). Each flight is analysed after its *departure* (first fix 3 km from the takeoff); before it, the pilot is in the *local phase* and counts as neither solo nor group. Analyses use loggers of at most 10 s; phases (climb, search, transition=glide) are Vilpellet's runs and exist on 1 Hz flights only (70% of the paragliders).
- **Contact.** i is in contact with j at t if j, at some t' with $|t' - t| \leq \delta = 2$ min, was within $\varepsilon_h = 500$ m horizontally and $\varepsilon_z = 300$ m vertically of i; when both glide, their headings differ by less than 45°. The delay accounts for pilots who follow the same route.
- **Pair interval.** A contact lasting at least $\tau^* = 38.1$ min, after joining gaps shorter than 2 min.
- **Configuration.** A fixed set of at least two flights that is one connected component of the pair intervals over a maximal interval; anyone joining or leaving starts a new one. **"n pilots" always means the size of the configuration at that instant.**
- **Episode.** A chain of configurations without break, each starting when the previous ends and sharing a member. **Block.** The flights tied by sharing episodes: the unit resampled for the error bars.

Parameter	Value	How it was set
δ, largest delay	2 min	*measured*: ratio of real to chance correspondences is 3.7 at 2 min (Fig. 1, top)
q, false discovery rate	0.1	choice: at most one contact in ten is chance
τ^*, shortest contact	38.1 min	*computed* from q and a null model (Fig. 1, bottom)
ε_h, ε_z, g	500 m, 300 m, 2 min	definitions (shared thermals within sight); no plateau on the data
R_{dep}	3 km	contacts at the takeoff say nothing

Table 1: Parameters of the grouping.

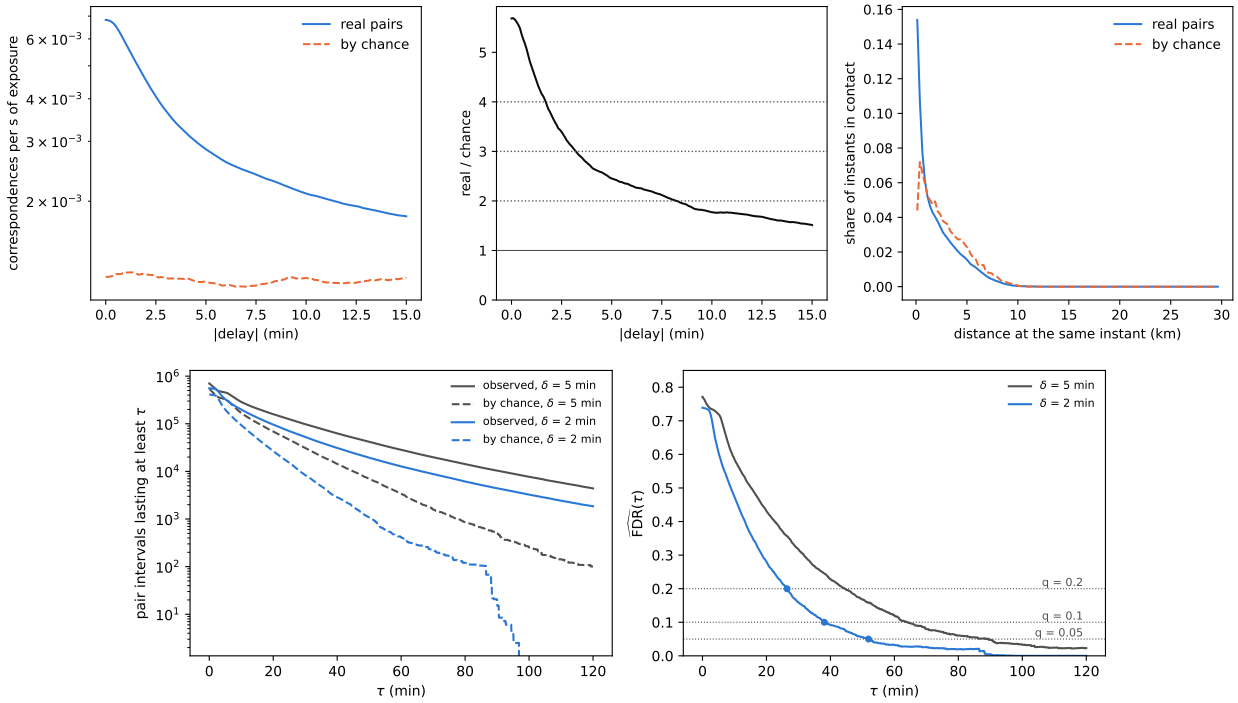


Figure 1: How the thresholds were set (tuning note). Top: correspondences per second at each delay, real pairs against chance (left), their ratio (middle), distance at the same instant (right). Bottom: real pair intervals lasting at least τ and those given by chance (left); estimated false discovery rate (right), with τ^* for $q = 0.05, 0.1, 0.2$.

Null model. Pairs of flights of one day with takeoffs within 10 km (at most 200 per day) are taken, and one flight of each pair is replaced by its own track shifted in time (at least $2\delta + 5$ s, so it cannot reproduce the real contact). Route, thermals and style are kept, the interaction is not. The contacts of the copies are chance: scaled to the flying time of the real pairs, day by day, they give $\widehat{\text{FDR}}(\tau)$; τ^* is where it falls to q. A contact is *shared air within sight*, not proof of interaction: pilots who use the same thermals at the same time without following each other survive the null model.

1.2 Populations, strata and statistics

- **S** solo flights; **G** group time (stretches in a configuration); **A** the *same flights alone* (stretches of group flights outside any configuration), the control with the same pilots, gliders, loggers and days; G_n the group time in configurations of n pilots.
- **TA-MSD.** Positions on a 10 s grid of the flight clock; per flight $\overline{\delta^2}(\tau) = \sum \|\mathbf{r}(t+\tau) - \mathbf{r}(t)\|^2$/#origins pooled over its stretches, ensemble MSD = mean over flights (equal weight); stretches of at least 1250 s, 21 lags from 10 to 1000 s. H: half the slope of $\log \text{MSD}$ against $\log \tau$ over all lags ($H = 1$ straight line). $H_{\text{loc}}(\tau)$: the same slope on the lags within ± 0.25 decades of τ (Appendix A).
- **Phase statistics.** Ratios of sums over runs (time in climb, searches that end in a climb, climb rate, glide speed, glide ratio; cross-country speed over 30 min). Five of twelve are drawn: the others are explained by them (Appendix C).
- **Strata.** *Zone*: cell of 25 km of the middle of a piece of track; *site-day*: flights of one day with takeoffs within 3 km; *hour*: local hour.
- **Error bars** are 5–95% bootstrap bands resampling whole *blocks* (never flights) and recomputing the statistic: blocks of the note for pooled statistics (1000 draws); blocks joining episodes and site-days for stratified ones (200 draws, same draws for all). They measure block sampling only, not the choice of q or the tolerances, and two bands that do not overlap are not a test; differences are computed draw by draw.

What the grouping gives. Of the 154 598 paraglider flights with a logger of at most 10 s, 33 501 (21.7%) fly in a group at some point; of the 6 039 hang-glider flights, 418 (6.9%). Group time is 12.8% of all paraglider flight time after the departure, and on average 45% of the time of a group flight. The 36 575 pair intervals last 40 to 96 min (10th to 90th percentile; median 52 min) and form 40 051 configurations in 17 169 episodes. Half of the episodes have two flights, 1% have more than 9, the largest has 53. By the size of the configuration, the paraglider group time is: 2 pilots 64%, 3 pilots 19%, 4 pilots 8%, 5-6 pilots 6%, 7+ pilots 3% (the largest configuration has 27 pilots). In contact, one pilot is behind the other by a median of 20 s (75% within 45 s).

2 Group or zone? Where the difference comes from

Question. Group time is more ballistic than solo flights at long lags. Is this the group, or the ground where groups fly (and where solo pilots would also progress in a straight line)? **Method.** Each piece of track is assigned to a *stratum* (zone = cell of 25 km of its middle; zone×month; zone×hour; site-day×hour). In every stratum with both populations the ensemble MSD of the solo flights S (and of the same flights alone A) is *reweighted to the mix of the group G over the strata*: S then flies, in the statistic, where and when G flies. We compare $H_{\mathrm{loc}}(\tau)$ over **every lag** and the fitted H. The share of G with a counterpart in its stratum is reported; "as is" is the same comparison without the reweighting, on the same strata and units (a flight counts once per cell it crosses).

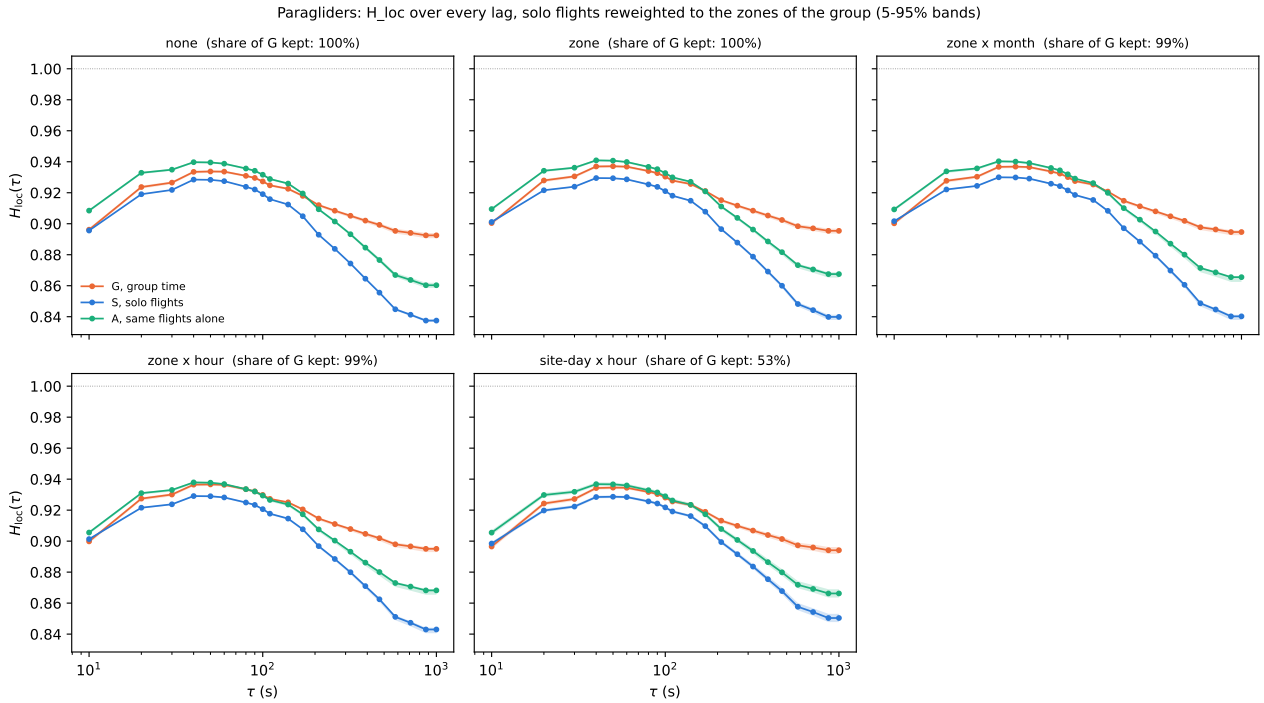


Figure 2: $H_{\mathrm{loc}}(\tau)$ of G (orange), of S reweighted to the strata of G (blue) and of A reweighted to them (green), 5–95% bands. First panel: all flights pooled (the raw comparison). "Share of G kept": flights of G with a solo counterpart in their stratum.

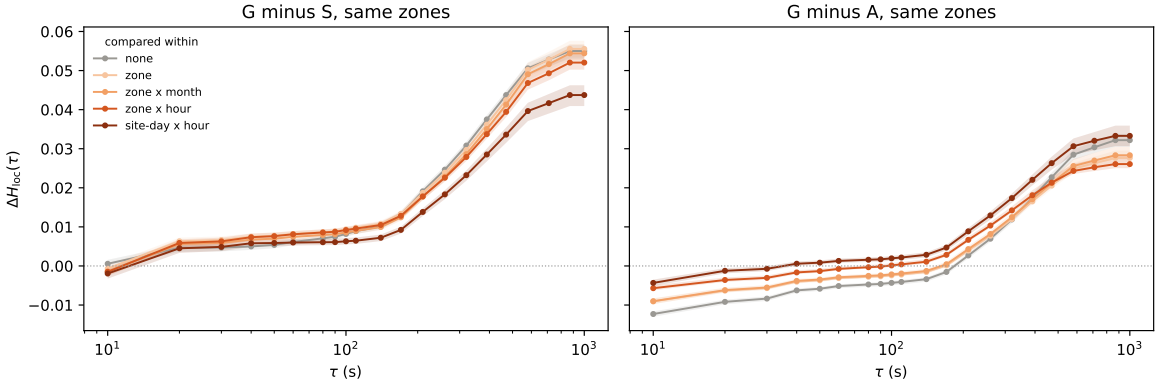


Figure 3: $\Delta H_{\mathrm{loc}}(\tau)$ over every lag: G minus S (left) and G minus A (right), the other population reweighted to the strata of G; one line per stratification, finer strata in darker orange.

compared within	G kept (%)	fitted H (10–1000 s)			ΔH			$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$		
		G	S | G	A | G	G − S as is	G − S reweighted	G − A reweighted	G − S as is	G − S reweighted	G − A reweighted
none	100	0.916	0.899	0.914	+0.017	+0.017 [+0.016, +0.018]	+0.002 [+0.001, +0.003]	+0.055	+0.055 [+0.053, +0.057]	+0.032 [+0.030, +0.034]
zone	100	0.919	0.902	0.916	+0.021	+0.018 [+0.017, +0.019]	+0.003 [+0.003, +0.004]	+0.059	+0.056 [+0.054, +0.058]	+0.028 [+0.026, +0.029]
zone × month	99	0.919	0.902	0.915	+0.022	+0.017 [+0.016, +0.018]	+0.003 [+0.003, +0.004]	+0.064	+0.054 [+0.053, +0.057]	+0.028 [+0.027, +0.031]
zone × hour	99	0.919	0.902	0.913	+0.022	+0.017 [+0.016, +0.018]	+0.005 [+0.005, +0.006]	+0.064	+0.052 [+0.050, +0.054]	+0.026 [+0.025, +0.028]
site-day × hour	53	0.917	0.904	0.913	+0.015	+0.013 [+0.012, +0.014]	+0.008 [+0.007, +0.009]	+0.051	+0.044 [+0.041, +0.046]	+0.033 [+0.031, +0.036]

Table 2: Fitted H and differences by stratification. "Reweighted": the other population reweighted to the strata of G (value, 5–95% band); "as is": the same strata without reweighting. $H_{\mathrm{S|G}}$: H of S reweighted to G.

Reading.

- **The gap opens above $\tau \approx 150$ s.** $\Delta H_{\mathrm{loc}}(\tau)$ between G and S is small below it (+0.005 to +0.011 from 20 to 140 s) and rises to +0.055 at 1000 s, above one climb–glide cycle: the group acts on how cycles are chained, not on a single glide or climb.
- **The zone does not explain it.** Reweighting S to the cells where G flies lowers $\Delta H_{\mathrm{loc}}(1000)$ by only 6% (+0.059 → +0.056) and the fitted ΔH by 15%; with the month, 15% and 23%; with the hour, 19% and 24%. At least 80% of the long-lag gap remains. Solo pilots fly where groups fly: 100% of G has a solo counterpart in its cell.
- **Same day, site and hour** (the finest strata, 53% of G kept): $\Delta H_{\mathrm{loc}}(1000)$ =+0.044 [+0.041, +0.046], 14% below its as-is value; fitted ΔH =+0.013 [+0.012, +0.014]. Solo flights on the same day, site and hour as a group still progress less straight over 1000 s.
- **Fitted H:** G 0.916, S 0.899 raw. S reweighted to the zones of G: 0.902; to zone×hour: 0.902; to site-day×hour: 0.904 (G 0.917). Reweighting raises the fitted H of S by a few thousandths and never brings it to that of G.
- **Against the same flights alone** (A): $\Delta H_{\mathrm{loc}}(1000)$ is +0.032 pooled and +0.028, +0.026, +0.033 in zone, zone×hour, site-day×hour strata: the same pilots in the same zones and hours progress straighter in a group. Below 150 s the sign depends on the hour: A is above G pooled and equal at the same site-day and hour, which is why the within-flight estimate of Section 4 must be read with the same-site-day-hour one.
- **Answer:** the difference between group and solo flights does not come from the zones where groups fly, nor from the month or hour at that place; it is reduced by about a fifth by the day, site and hour together, and the rest stays with the pilot and the day fixed. It is not a zone effect.

Limits. Cells are 25 km, from the middle of a piece; the terrain inside a cell (ridge, plain) is not controlled, and a group may choose better air within it: this is part of what we call the group effect. Paragliders only.

3 Group or moment? The same flight before, during and after

Question. Pilots may choose to fly together in the moments and places where the progress is straight anyway. If so, the same flight would already be straight just before the group forms and just after it breaks. **Method.** Every stretch of group time lasting at least 50 min is an *event* with up to four windows of 50 min of the flight clock: *before* (the 50 min preceding it, in the solo time of the same flight), *start* and *end* of the group (its first and last 50 min), *after*. Before and after need a solo stretch of at least that length next to the group, so a group formed soon after the departure has no *before*. For two windows the ensemble MSD is taken on the events that have both, and compared over **every lag**. **Controls:** two consecutive windows of the same length with no change of state, drawn in the solo time of the group flights (A) and in solo flights (S), which measure the drift of the flight between consecutive windows.

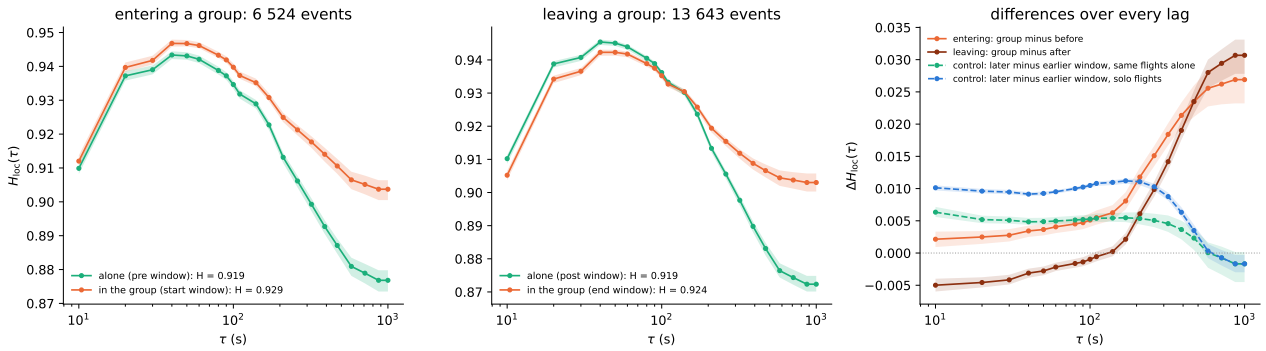


Figure 4: $H_{\text{loc}}(\tau)$ of the same flights alone (green) and in the group (orange), entering (left) and leaving (middle), 5–95% bands. Right: the differences, with the two controls (later minus earlier window; dashed).

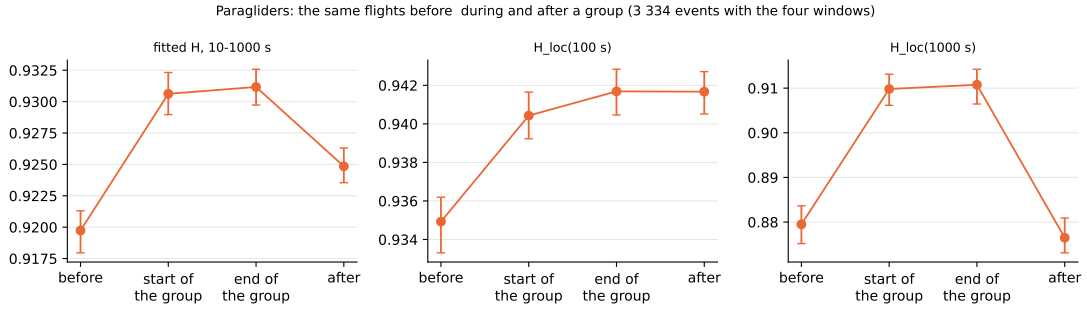


Figure 5: The four windows on the events that have all of them. Start and end overlap when the group lasts less than 2×50 min.

comparison	events	ΔH	$\Delta H_{\text{loc}}(100)$	$\Delta H_{\text{loc}}(1000)$
entering: start − before	6 524	+0.010 [+0.009, +0.011]	+0.005 [+0.004, +0.006]	+0.027 [+0.023, +0.031]
leaving: end − after	13 643	+0.005 [+0.004, +0.006]	−0.001 [−0.002, −0.000]	+0.031 [+0.028, +0.033]
after − before	6 320	+0.004 [+0.003, +0.005]	+0.006 [+0.005, +0.007]	−0.003 [−0.007, +0.001]
control A: later − earlier	16 998	+0.005 [+0.004, +0.005]	+0.005 [+0.005, +0.006]	−0.002 [−0.005, +0.000]
control S: later − earlier	69 648	+0.009 [+0.008, +0.009]	+0.010 [+0.010, +0.011]	−0.002 [−0.003, −0.000]

Table 3: Differences of the fitted H and of H_{loc} at 100 and 1000 s (value, 5–95% band) between windows. Controls: two consecutive windows with no change of state, in the solo time of the group flights (A) and in solo flights (S). Events: those with both windows.

comparison	events	time in climb (points)	searches to climb (points)	climb rate (m/s)	glide speed (km/h)	height lost per km (m/km)
entering: start − before	4 786	+0.7 [+0.3, +1.1]	+0.0 [−1.1, +1.0]	+0.046 [+0.034, +0.062]	+0.92 [+0.69, +1.15]	−2.3 [−3.5, −1.3]
leaving: end − after	10 097	+1.5 [+1.2, +1.9]	+1.2 [+0.4, +1.9]	−0.007 [−0.017, +0.003]	+0.52 [+0.35, +0.64]	−10.9 [−11.8, −10.0]
after − before	4 718	−1.4 [−1.8, −1.0]	−2.8 [−4.0, −1.6]	+0.043 [+0.024, +0.059]	+0.33 [+0.12, +0.55]	+9.1 [+7.8, +10.4]
control A: later − earlier	12 568	−1.1 [−1.3, −0.9]	−1.4 [−2.0, −0.8]	+0.018 [+0.009, +0.027]	+0.07 [−0.05, +0.20]	+3.3 [+2.5, +4.2]
control S: later − earlier	49 092	−1.6 [−1.7, −1.5]	−2.3 [−2.7, −2.0]	+0.011 [+0.008, +0.015]	+0.04 [−0.03, +0.10]	+4.9 [+4.5, +5.3]

Table 4: The phase statistics of the same comparisons (differences with 5–95% band); runs and glides are placed in the window that holds their middle.

Reading.

- **Over 1000 s the same flights are straighter in the group, and straightness follows the group.** $\Delta H_{\text{loc}}(1000)$ is +0.027 [+0.023, +0.031] entering and +0.031 [+0.028, +0.033] leaving, against −0.002 and −0.002 between consecutive windows with no change of state. The gap opens above about 150 s (Fig. 4).
- **It does not belong to the pilot or to the day.** The windows before and after are alike: $\Delta H_{\text{loc}}(1000)$ after minus before is −0.003 [−0.007, +0.001]. On the 3 334 events with all four windows $H_{\text{loc}}(1000)$ is 0.879 before, 0.910 and

0.911 in the group, 0.876 after (Fig. 5). The step is of the order of the within-flight estimate of Section 4 (+0.037).

- **Nothing within one glide or climb:** at 100 s the changes are +0.005 entering and −0.001 leaving, inside the drift of the controls (+0.005, +0.010). The fitted H gives +0.010 and +0.005 against +0.005 and +0.009: it does not separate from the drift, because its slope is dominated by the lags below 150 s.
- **Glides are faster and flatter in the group, in both directions:** glide speed +0.92 km/h entering and +0.52 leaving, height lost per km −2.3 and −10.9 m/km, against +0.04 and +0.07 km/h, +3.3 and +4.9 m/km for the controls (leaving is group minus alone, the controls are later minus earlier). Time in climb, searches that end in a climb and climb rate move with the window by the size of the drift and with a sign that depends on the direction: no robust change.

Limits. The comparison fixes the pilot, the glider, the day and the slow drift of the flight, not the moment: what brings pilots together (a convergence line, a ridge, a common route) may last as long as the group and straighten the progress by itself, and two pilots must move alike to stay within 500 m for 50 min. Events are long groups with at least 50 min of solo flight beside them (6 524 entries, 13 643 exits); phases on 1 Hz flights only. The same test with 21-min windows is in Appendix G.

4 Other statistics: what the group changes

	solo flights S	group time G	same flights alone A
flights (TA-MSD)	119 129	33 435	29 611
H (10–1000 s)	0.899 [0.898, 0.899]	0.916 [0.915, 0.917]	0.914 [0.913, 0.915]
$H_{\mathrm{loc}}(100\ \mathrm{s})$	0.919 [0.919, 0.919]	0.927 [0.927, 0.928]	0.932 [0.931, 0.932]
$H_{\mathrm{loc}}(1000\ \mathrm{s})$	0.837 [0.837, 0.838]	0.893 [0.891, 0.894]	0.860 [0.859, 0.862]
time in climb (%)	30.8 [30.7, 30.9]	33.0 [32.8, 33.2]	29.1 [28.9, 29.2]
searches ending in a climb (%)	43.3 [43.2, 43.4]	49.5 [49.2, 49.7]	45.9 [45.7, 46.1]
climb rate (m/s)	1.22 [1.22, 1.22]	1.26 [1.25, 1.27]	1.33 [1.32, 1.34]
glide speed (km/h)	30.1 [30.1, 30.2]	33.3 [33.1, 33.5]	32.8 [32.6, 32.9]
glide ratio	10.4 [10.4, 10.4]	12.9 [12.8, 13.0]	11.3 [11.2, 11.4]
cross-country speed (km/h)	17.2 [17.2, 17.3]	20.0 [19.8, 20.3]	20.0 [19.8, 20.1]

Table 5: Pooled statistics of S, G and A (paragliders), with 5–95% bands (blocks of the note). TA-MSD on loggers of at most 10 s, phases on 1 Hz flights.

Design. Five contrasts, each computed inside strata and averaged over those with both populations (weights $n_P n_Q/(n_P+n_Q)$, n the flights of each side): G − S pooled; G − S and A − S (*selection*) at the same site-day and hour; G − A within the flight and at the same site-day and hour (the *group effect*, estimated in two ways that fail in opposite ways: the first compares a flight with itself at another time of day, the second compares pilots at the same time). Where the last two agree, the effect belongs to the group.

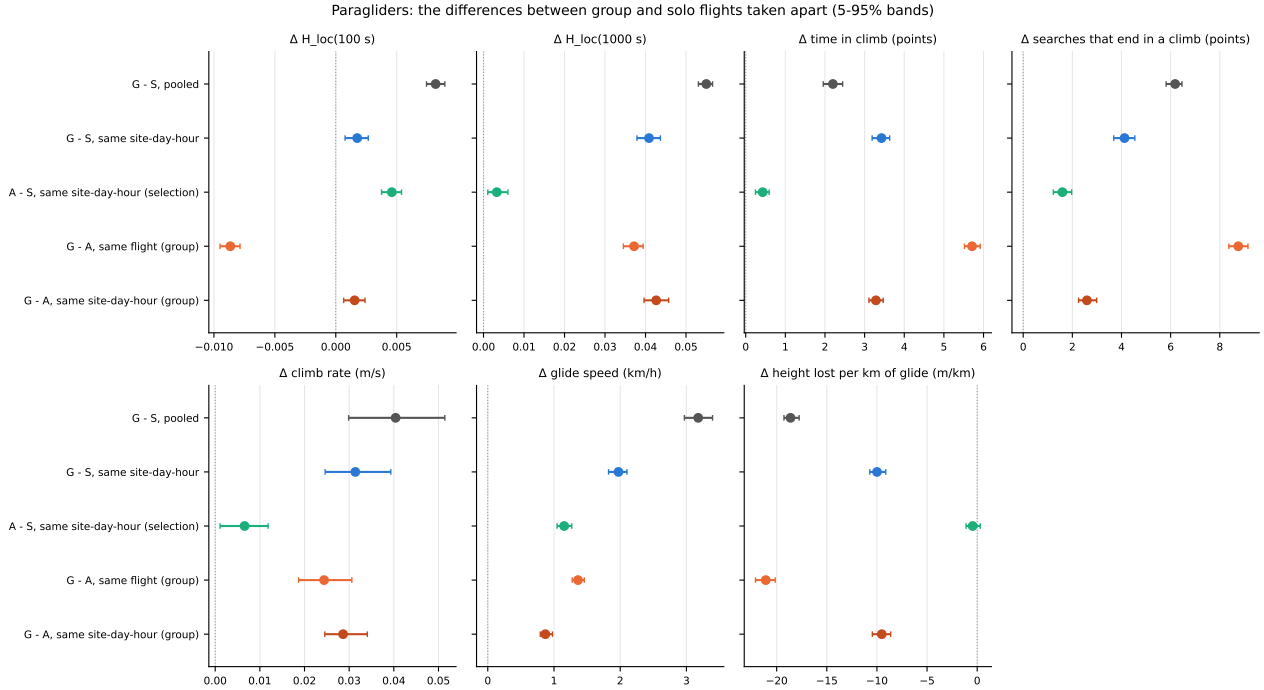


Figure 6: Each statistic (panel) for the five contrasts (rows), with 5–95% bands. Shares of time in points; lower height lost per km is a better glide.

Δ	G − S pooled	G − S same s-d-h	A − S same s-d-h	G − A same flight	G − A same s-d-h
$H_{\mathrm{loc}}(100\ \mathrm{s})$	+0.008 [+0.007, +0.009]	+0.002 [+0.001, +0.003]	+0.005 [+0.004, +0.005]	−0.009 [−0.009, −0.008]	+0.002 [+0.001, +0.002]
$H_{\mathrm{loc}}(1000\ \mathrm{s})$	+0.055 [+0.053, +0.057]	+0.041 [+0.038, +0.044]	+0.003 [+0.001, +0.006]	+0.037 [+0.035, +0.039]	+0.043 [+0.040, +0.046]
time in climb (%)	+2.20 [+1.96, +2.45]	+3.43 [+3.19, +3.63]	+0.43 [+0.25, +0.59]	+5.71 [+5.52, +5.92]	+3.28 [+3.11, +3.47]
searches to climb (%)	+6.18 [+5.82, +6.46]	+4.12 [+3.68, +4.54]	+1.60 [+1.23, +1.97]	+8.75 [+8.37, +9.15]	+2.59 [+2.25, +2.99]
climb rate (m/s)	+0.040 [+0.030, +0.051]	+0.031 [+0.025, +0.039]	+0.007 [+0.001, +0.012]	+0.024 [+0.019, +0.031]	+0.029 [+0.025, +0.034]
glide speed (km/h)	+3.18 [+2.97, +3.39]	+1.97 [+1.82, +2.10]	+1.15 [+1.05, +1.27]	+1.36 [+1.27, +1.46]	+0.87 [+0.79, +0.98]
height lost per km (m/km)	−18.6 [−19.3, −17.8]	−10.0 [−10.7, −9.1]	−0.5 [−1.1, +0.3]	−21.1 [−22.1, −20.1]	−9.5 [−10.5, −8.6]
cross-country speed (km/h)	+2.81 [+2.59, +3.01]	+2.25 [+2.09, +2.39]	+1.31 [+1.21, +1.43]	+0.70 [+0.60, +0.78]	+1.29 [+1.16, +1.41]

Table 6: The contrasts of Figure 6: difference and 5–95% band (s-d-h: same site-day and hour).

Reading.

- **Straighter progress over 1000 s** survives every control: $\Delta H_{\mathrm{loc}}(1000)$ is +0.055 pooled, +0.037 within the flight, +0.043 at the same site-day and hour; selection only +0.003.
- **Nothing within one phase (100 s):** −0.009 within the flight, +0.002 at the same place and hour; the pooled +0.008 was selection and composition.
- **More time climbing** (+5.71, +3.28 points) and **more searches that end in a climb** (+8.75, +2.59 points).
- **Climb rate is higher in a group** (+0.024, +0.029 m/s), not lower as in the pooled table: a composition effect (pooled ratios weight flights by their time in each population).
- **Faster and flatter glides:** +1.36, +0.87 km/h and −21.1, −9.5 m/km less height lost per km; selection adds +1.15 km/h of glide speed (group pilots glide faster even alone).
- **Cross-country speed** is higher in a group once stratified: +0.70, +1.29 km/h.

4.1 Size of the configuration

The group time is cut where the configuration changes: every stretch has a fixed number of pilots n (2, 3, 4, 5–6, 7+); a flight can contribute to several sizes. Two contrasts by size: each size minus the same flights alone (within the flight), and each size minus configurations of 2 pilots (same site-day and hour: does the size matter beyond a pair?).

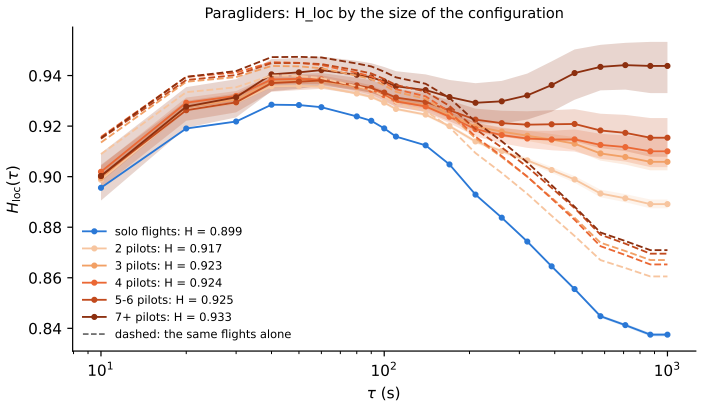


Figure 7: $H_{\mathrm{loc}}(\tau)$ over every lag by the size of the configuration (orange, darker with more pilots), S (blue) and, dashed, the same flights alone for each size.

		H		$H_{\mathrm{loc}}(1000\ \mathrm{s})$						
pilots	flights	G_n	alone	G_n	alone	time in climb (%)	searches to climb (%)	climb rate (m/s)	glide speed (km/h)	glide ratio
solo	119 129	0.899		0.837		30.8	43.3	1.22	30.1	10.4
2	26 411	0.917	0.914	0.889	0.861	32.7	49.0	1.27	33.2	12.5
3	9 480	0.923	0.919	0.906	0.867	33.6	49.9	1.26	33.5	13.1
4	4 073	0.924	0.920	0.910	0.865	33.6	51.0	1.25	33.5	13.7
5–6	2 686	0.925	0.922	0.915	0.870	33.1	50.1	1.23	33.1	14.1
7+	1 142	0.933	0.924	0.944	0.871	34.6	51.7	1.20	34.6	15.9

Table 7: Paragliders, pooled by the size of the configuration ("alone": the same flights alone).

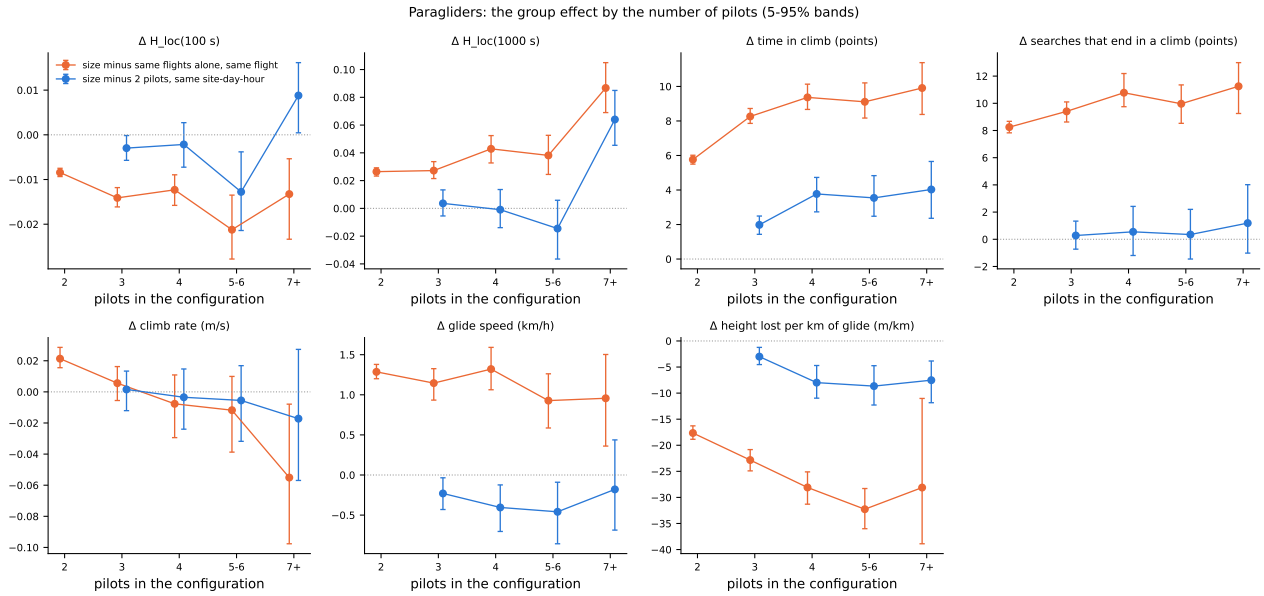


Figure 8: Orange: $G_n - A$ within the flight. Blue: $G_n - G_2$ at the same site-day and hour.

Reading. Pooled, H and H_{loc} rise with n while the same flights alone stay flat (Fig. 7, Table 7). Within the flight the straight-progress effect grows with n (+0.026 with 2 pilots, +0.087 with 7+), but at the same site-day and hour configurations of 3 to 6 pilots do not differ from pairs ($\Delta H_{\mathrm{loc}}(1000)$ between −0.015 and +0.004, bands across zero): the rise of the pooled curves belongs to the days and sites where large groups form. Only 7+ stands apart (+0.064 [+0.045, +0.085], cross-country speed +7.0 km/h), a rare configuration (3% of group time) of unknown nature. Time in climb and the flatter glide do grow with n at the same place and hour (+2.0 to +4.0 points; −3 to −9 m/km).

4.2 Leader and follower

In configurations of two pilots the signed median delay of the pair interval says who passes first (leader: the other passes at least 15 s later; 35% abreast, median delay 20 s). The two are compared within the configuration (Appendix H). How groups form and break, whether the same pilots lead, and when groups fly are in the report *Group structure*.

Reading. The pilot ahead flies a little faster (glide speed +0.86 [+0.74, +0.99] km/h, climb rate +0.021 [+0.014, +0.028] m/s) but does not find thermals more often (searches ending in a climb −0.01 [−1.03, +0.83] points, height lost per km +1.6 [−0.9, +4.5] m/km): within a pair the follower shows no sign of using the leader to find the next thermal. Group time is concentrated around midday and its time alone falls earlier and later (*Group structure*): this is why G − A within the flight must be read with the same-site-day-hour estimate.

5 Conclusions and limits

Supported

- the progress over a quarter of an hour is straighter in a group, with the pilot, the day, the site and the hour fixed;
- no effect within one glide or climb (100 s) at the same hour;
- on the same flight the straighter progress over 1000 s starts with the group and stops with it (before $\approx$ after);
- more time climbing, more searches that end in a climb, faster and flatter glides;
- climb rate and cross-country speed are *higher* in a group once the composition is controlled (the pooled readings reversed);
- selection of pilots is small for the motion;
- the pilot ahead flies a little faster but finds no more thermals than the follower.

Not supported

- that the group *causes* the effect: the strata and the same-flight windows fix pilot, day, site, hour, zone and the drift of the flight, not what brings pilots together (a line, a ridge, a route) nor the coordination needed to stay in contact;
- that larger groups straighten the progress, up to six pilots;
- that a contact means interaction (shared air within sight);
- the glide statistics are partly by definition (aligned headings in glide);
- phases exist on 1 Hz flights only; the same analyses on the xc-dynamics tracks are open.

A How H and H_{loc} are computed

$\mathrm{MSD}(\tau) \propto \tau^{2H}$, so $\log_{10}\mathrm{MSD} = 2H\log_{10}\tau + c$. H is half the slope of the least-squares line over all 21 lags (equal weight per lag). $H_{\mathrm{loc}}(\tau_c)$ is half the slope of the line fitted on the lags with $|\log_{10}\tau - \log_{10}\tau_c| \leq 0.25$ (a factor 1.78 on each side), equal weight; where fewer than 3 lags lie in the window, the 3 nearest are used. It is a windowed regression, not a pointwise derivative. ΔH and ΔH_{loc} of two curves are the differences of the fitted values; for a stratified average they are the same regression on the averaged $\Delta\log\mathrm{MSD}$ (linear: slope of a difference = difference of slopes). Bands: the same computation on each bootstrap draw, percentiles lag by lag.

τ (s)	lags used	τ (s)	lags used	τ (s)	lags used
10	10–30 (3 nearest)	90	60–140	320	210–470
20	20–40 (3 nearest)	100	60–170	390	260–580
30	20–50	110	80–170	470	320–710
40	30–60	140	80–210	580	390–1000
50	30–80	170	100–260	710	470–1000
60	40–100	210	140–320	870	580–1000
80	50–140	260	170–390	1000	580–1000

Table 8: Lags of the window of each H_{loc} value. At 10 and 20 s the window is not centred; at 870 and 1000 s it is cut at 1000 s and the two values use the same lags, so the end of the curve flattens by construction. The window resolves nothing narrower than about half a decade.

Phase MSD figures (not in this report) use a different local exponent: the finite-difference derivative of $\log\mathrm{MSD}$ against $\log\Delta$ on 100 lags, without the factor $1/2$ (about 2 for ballistic motion).

Ratios and the glide ratio. Ratios are sums of numerators over sums of denominators over the runs of a population (a run goes to the stretch that holds its middle). In a small stratum a glide in rising air has almost no height lost, the glide ratio diverges and its average over strata is unstable; stratified contrasts use the *height lost per km of glide*, the inverse. Pooled tables keep the glide ratio.

B Checks of the grouping and of the viewer

- **Code against the methodological note.** Every phase was compared with its code (grid, departure, candidate filter, scan of delays, contact, gaps, signed delay, null model, configurations, episodes, membership, blocks): all implemented as written, with four differences of text, **now corrected in red in the note**: (1) the phases of the run are Vilpellet's, on 1 Hz flights, not the hidden Markov model of xc-lab, so the heading condition acts only between two such flights; (2) headings are in the east and north of the first fix of each flight, not of the fix itself (a rotation of about 3° at 300 km, against 45°); (3) a contact is kept only from the later departure to the earlier landing (at most δ lost at the ends); (4) instants in a gap of the track are labelled solo in the tables.
- **Properties of the tables** (chosen run): every pair interval lasts at least $\tau^* = 2285$ s and lies where both flights fly after departure; every configuration has at least two members; membership covers each flight from takeoff to landing without holes; on 3 000 flights the group time equals the union of its pair intervals; on 3 000 configurations the members are connected at the midpoint and no active interval leaves; on 2 000 episodes every configuration but the first starts when another ends and shares a member; every episode lies in one block. The counts of the note (154 598 and 6 039 flights; 21.7% and 6.9% in a group) are reproduced.
- **Viewer.** The *Group flights* tab reads `groups/` of its data folder: its seven tables are **byte-identical** to those of the chosen run; the tracks it draws (`SSD_DATA`) have the same fixes, times, positions and trimmed starts as those the grouping read (`SSD_DISANTE`).
- **Analyses.** Tests on synthetic flights for every step, and on every number of the README of the analyses; the decomposition is checked on a synthetic case where the day raises a statistic by 10 and the group by 2: pooled 10, stratified 2.

C Which statistics are drawn

Unit: a population on a site-day with at least 25 flights (119 units). A statistic is redundant when the others explain most of it across units (R^2 of its linear regression on them); backward elimination drops the most explained until every $R^2 \leq 0.5$. The four shares of time sum to 100%, so one of them goes first.

statistic	status	R^2 on the five drawn
time in climb (%)	drawn	0.23
searches that end in a climb (%)	drawn	0.41
climb rate (m/s)	drawn	0.20
glide speed (km/h)	drawn	0.26
glide ratio	drawn	0.38
time in transition (%)	dropped, number 1, R^2 1.00 when dropped	0.83
height gained per climb (m)	dropped, number 2, R^2 0.98 when dropped	0.91
climbs that start after a search (%)	dropped, number 3, R^2 0.93 when dropped	0.77
cross-country speed over 30 min (km/h)	dropped, number 4, R^2 0.93 when dropped	0.90
time in a search in vain (%)	dropped, number 5, R^2 0.88 when dropped	0.71
climbs per hour	dropped, number 6, R^2 0.72 when dropped	0.71
time in a search that ends in a climb (%)	dropped, number 7, R^2 0.65 when dropped	0.65

Table 9: Five statistics remain; the others are explained by them.

D Pooled curves, hang gliders and sizes

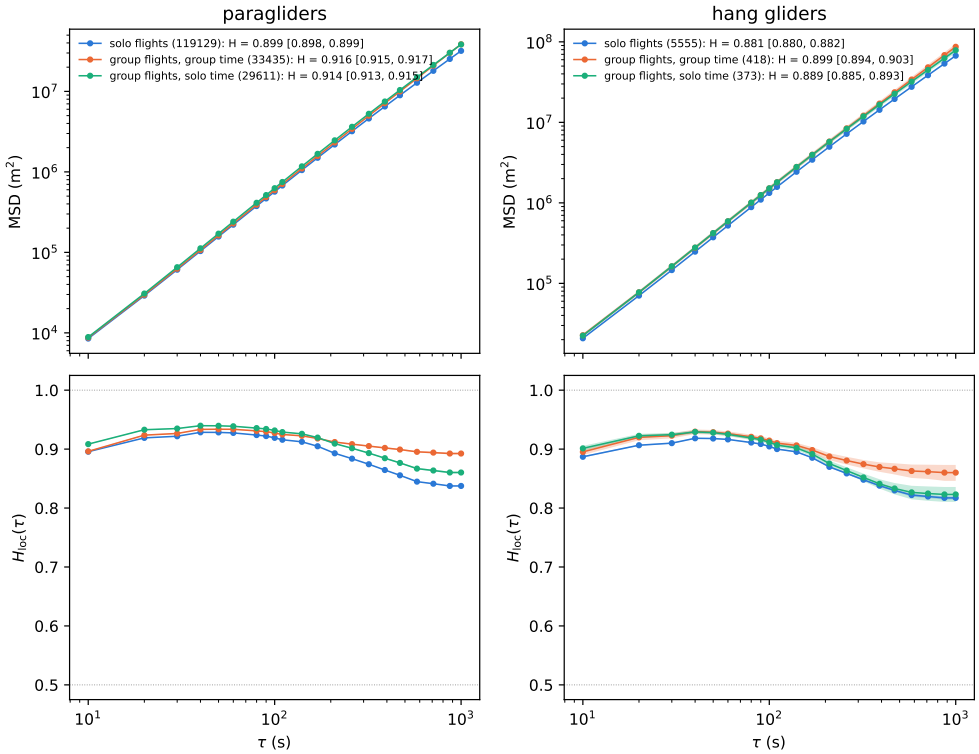


Figure 9: TA-MSD and H_{loc} of S, G and A, paragliders and hang gliders (hang gliders: 418 group flights, wide bands at long lags).

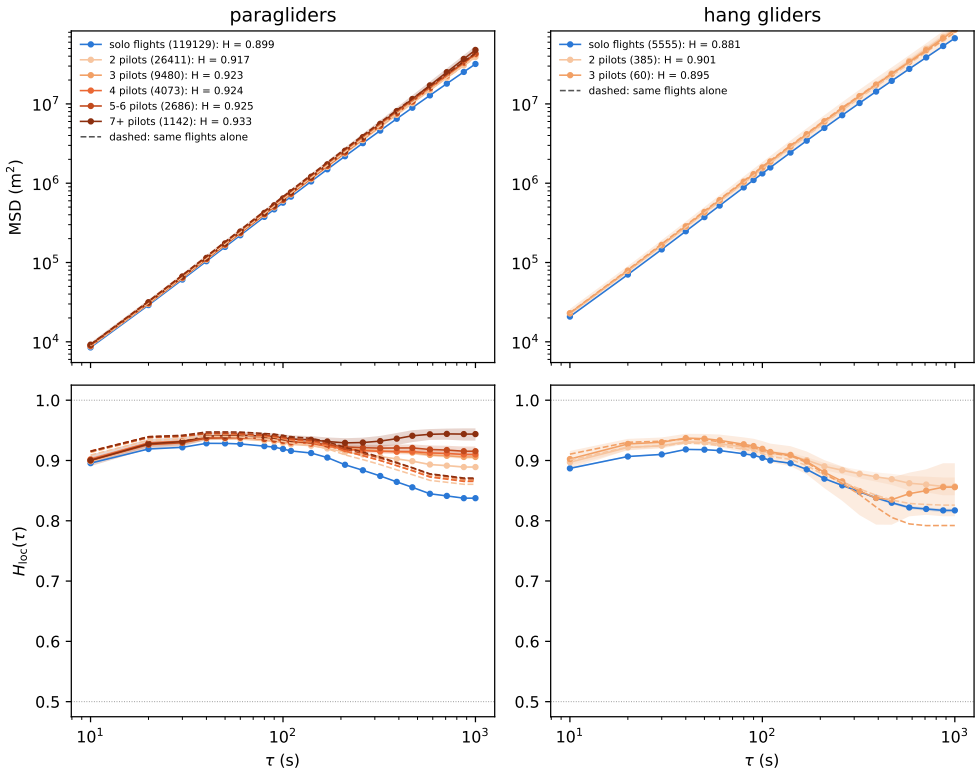


Figure 10: By the size of the configuration (orange, darker with more pilots), S (blue) and, dashed, the same flights alone for each size.

E Single site-days

Six site-days chosen by a fixed rule (most sizes with 15 flights or more, then most flights, at most two per takeoff site). Bands resample flights (too narrow: flights of one group are not independent).

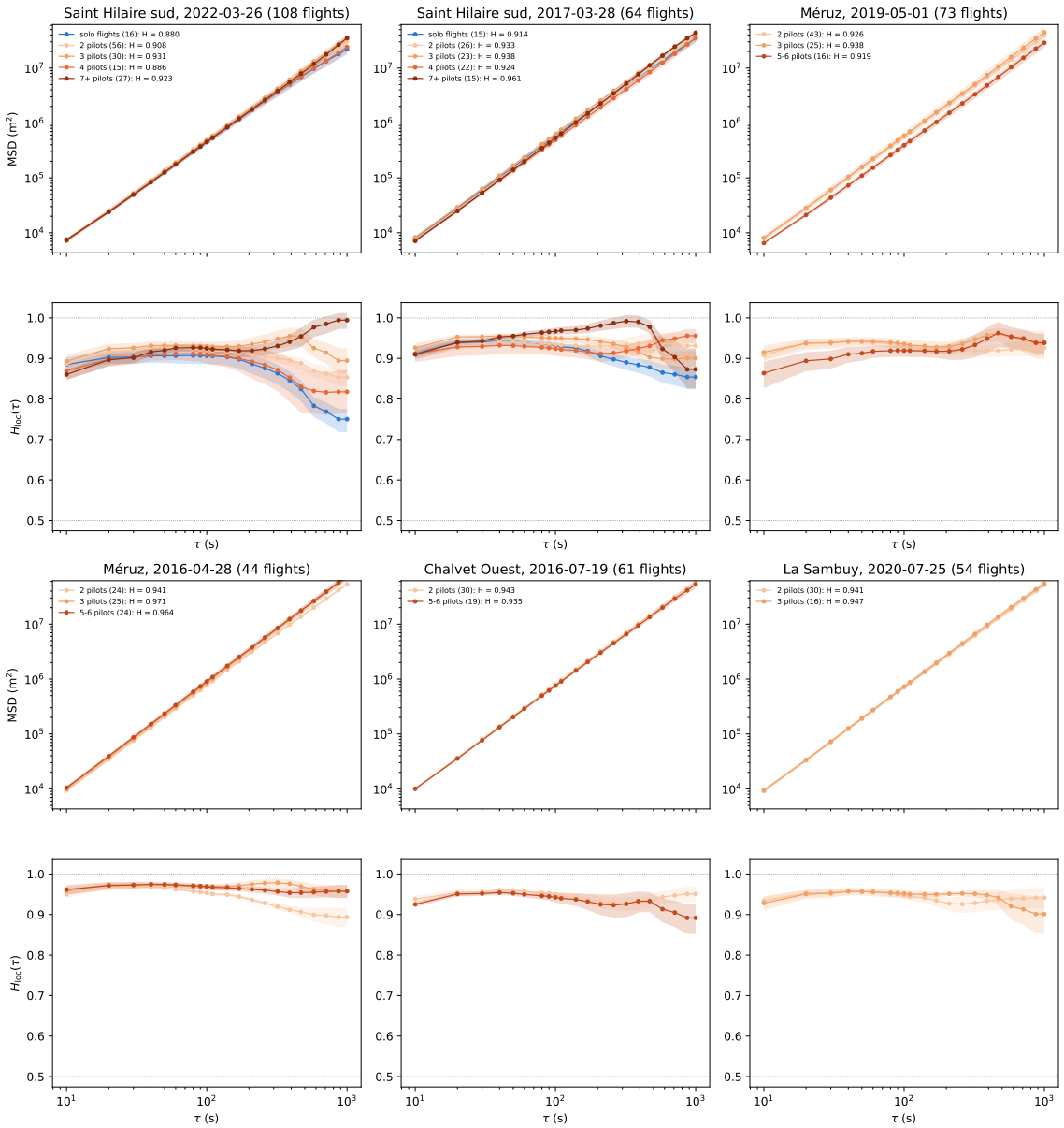


Figure 11: TA-MSD and H_{loc} on six site-days by the size of the configuration, with the solo flights of the same site-day where they reach 15 flights.

site-day	solo	2	3	4	5–6	7+
Saint Hilaire sud, 2022-03-26	0.750	0.853	0.894	0.818		0.994
Saint Hilaire sud, 2017-03-28	0.854	0.931	0.901	0.956		0.873
Méruz, 2019-05-01		0.926	0.939		0.938	
Méruz, 2016-04-28		0.894	0.958		0.957	
Chalvet Ouest, 2016-07-19		0.951			0.892	
La Sambuy, 2020-07-25		0.941	0.901			

Table 10: $H_{\text{loc}}(1000\text{ s})$ by size on each site-day (empty: fewer than 15 flights). Differences between days exceed those between sizes (the climb rate goes from 0.79 to 2.11 m/s across the six days).

F Robustness to the threshold

With $q = 0.2$ contacts need to last only 26.4 min: more and shorter groups, up to one contact in five chance.

	$q = 0.1$, $\tau^* = 38.1$ min	$q = 0.2$, $\tau^* = 26.4$ min
group flights (TA-MSD)	33 435	48 777
$H_{\mathrm{loc}}(1000\ \mathrm{s})$, S	0.837	0.836
$H_{\mathrm{loc}}(1000\ \mathrm{s})$, A	0.860	0.854
$H_{\mathrm{loc}}(1000\ \mathrm{s})$, G	0.893	0.892
$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$, G − A same flight	+0.037 [+0.035, +0.039]	+0.032 [+0.029, +0.035]
$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$, G − A same s-d-h	+0.043 [+0.040, +0.046]	+0.042 [+0.040, +0.044]
$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$, A − S same s-d-h	+0.003 [+0.001, +0.006]	+0.002 [−0.001, +0.005]
$\Delta H_{\mathrm{loc}}(100\ \mathrm{s})$, G − A same flight	−0.009 [−0.009, −0.008]	−0.016 [−0.017, −0.015]
$\Delta H_{\mathrm{loc}}(100\ \mathrm{s})$, G − A same s-d-h	+0.002 [+0.001, +0.002]	−0.004 [−0.005, −0.003]
Δ searches to climb, G − A same s-d-h (points)	+2.6 [+2.3, +3.0]	+3.4 [+3.0, +3.7]
Δ climb rate, G − A same s-d-h (m/s)	+0.029 [+0.025, +0.034]	+0.024 [+0.020, +0.029]
Δ height lost per km, G − A same s-d-h (m/km)	−9.5 [−10.5, −8.6]	−11.5 [−12.2, −10.9]
Δ cross-country speed, G − A same s-d-h (km/h)	+1.29 [+1.16, +1.41]	+1.14 [+1.01, +1.30]

Table 11: The main results with the two thresholds (s-d-h: same site-day and hour). Every conclusion keeps its sign; at 100 s both estimates of the group effect are slightly negative with $q = 0.2$: no robust effect within one phase.

G Before, during and after with short windows

Same test as Section 3 with windows of 21 min, the shortest a piece of the TA-MSD allows. Groups of at least 21 min enter: more events, but at 1000 s a window holds about 26 origins, so H_{loc} over the longest lags is noisy and its drift between consecutive windows is large (controls below). The fitted H, dominated by short lags, shows no change beyond the drift; the phase statistics at entry depend on the window (time in climb +3.3 points, glide speed +0.63 km/h here, against +0.7 and +0.92 with 50 min).

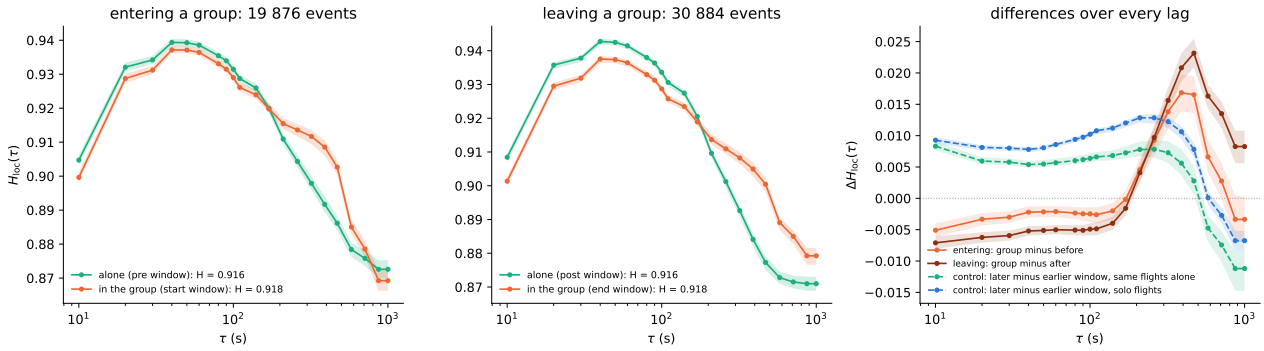


Figure 12: As Fig. 4, windows of 21 min.

comparison	events	ΔH	$\Delta H_{\mathrm{loc}}(100)$	$\Delta H_{\mathrm{loc}}(1000)$
entering: start − before	19 876	+0.002 [+0.001, +0.004]	−0.002 [−0.004, −0.001]	−0.003 [−0.007, +0.000]
leaving: end − after	30 884	+0.002 [+0.001, +0.003]	−0.005 [−0.006, −0.004]	+0.008 [+0.006, +0.011]
after − before	14 482	+0.004 [+0.003, +0.005]	+0.006 [+0.005, +0.008]	+0.002 [−0.001, +0.006]
control A: later − earlier	33 658	+0.006 [+0.005, +0.006]	+0.006 [+0.006, +0.007]	−0.011 [−0.015, −0.008]
control S: later − earlier	107 324	+0.009 [+0.009, +0.010]	+0.010 [+0.010, +0.011]	−0.007 [−0.008, −0.005]

Table 12: As Table 3, windows of 21 min.

comparison	events	time in climb (points)	searches to climb (points)	climb rate (m/s)	glide speed (km/h)	height lost per km (m/km)
entering: start − before	14 798	+3.3 [+2.9, +3.8]	+1.2 [+0.3, +2.2]	+0.043 [+0.035, +0.053]	+0.63 [+0.45, +0.77]	−8.9 [−12.1, −6.2]
leaving: end − after	23 014	+2.7 [+2.3, +3.1]	+0.9 [+0.2, +1.6]	+0.008 [+0.000, +0.018]	+0.54 [+0.42, +0.67]	−13.1 [−14.5, −11.5]
after − before	10 898	−1.5 [−2.0, −1.1]	−4.1 [−5.2, −2.9]	+0.017 [−0.001, +0.030]	+0.17 [−0.01, +0.35]	+9.6 [+5.7, +12.9]
control A: later − earlier	25 121	−2.1 [−2.4, −1.8]	−2.8 [−3.5, −2.0]	−0.001 [−0.009, +0.006]	−0.29 [−0.40, −0.15]	+3.4 [+1.9, +5.2]
control S: later − earlier	75 984	−2.0 [−2.2, −1.8]	−3.0 [−3.5, −2.7]	−0.012 [−0.017, −0.009]	−0.31 [−0.37, −0.24]	+6.2 [+5.0, +7.5]

Table 13: As Table 4, windows of 21 min.

H Leader minus follower

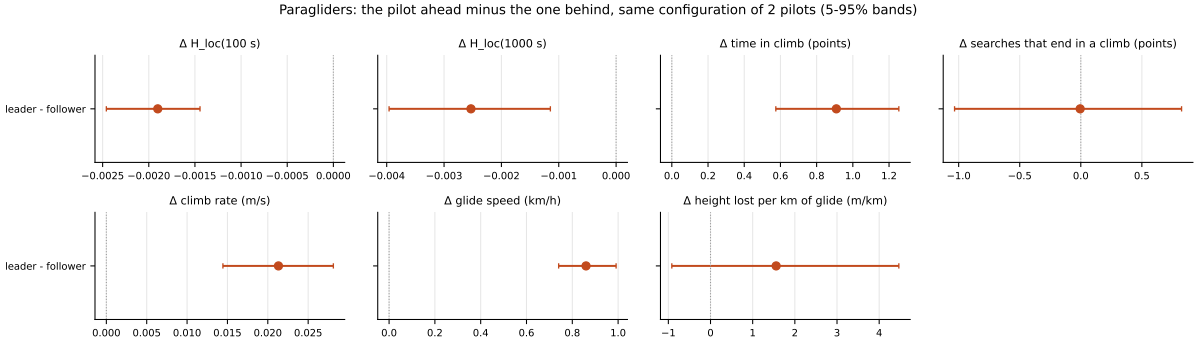


Figure 13: Leader minus follower, same configuration of two pilots.

Δ	leader − follower
$H_{\text{loc}}(100\ \text{s})$	−0.002 [−0.002, −0.001]
$H_{\text{loc}}(1000\ \text{s})$	−0.003 [−0.004, −0.001]
time in climb (%)	+0.91 [+0.57, +1.25]
searches ending in a climb (%)	−0.01 [−1.03, +0.83]
climb rate (m/s)	+0.021 [+0.014, +0.028]
glide speed (km/h)	+0.86 [+0.74, +0.99]
height lost per km of glide (m/km)	+1.6 [−0.9, +4.5]
cross-country speed (km/h)	+0.14 [+0.11, +0.16]

Table 14: Leader minus follower, same configuration of two pilots (difference with 5–95% band).

I Reproducing the report

From `xc-lab/analysis/groups/group-identification`, SSD_DISANTE mounted:

```
uv run python tamsd.py; uv run python phase_msd.py; uv run python survival.py
uv run python budget.py; uv run python performance.py; uv run python blocks.py
uv run python redundancy.py; uv run python size.py; uv run python decompose.py
uv run python zones.py; uv run python prepost.py; uv run python prepost.py --window 1260
uv run python mechanisms.py; uv run python report/facts.py
uv run python report/tables.py; cd report; latexmk -pdf group_vs_solo.tex
```

Heavy pieces stay on SSD_DISANTE, in the `analyses/` folder of each run; figures and JSON in `out/<run>/`.